

Optimal Control Using Koopman Operator

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All of the simulations of this work have their MATLAB codes attached to the report.

Abstract: This project addresses the optimal control of nonlinear system using Koopman operator. First, a linear representation of the nonlinear system is derived via Koopman eigenfunction. Then, an LQR controller is designed for the linear system. This method is tested on a Duffing Oscillator system with both Data-driven and analytical perspective.

Index Terms: Nonlinear Control, Koopman Operator, Machine-Learning Control, Linear Quadratic Regulator.

1. Introduction

There are powerful methods for control of the linear systems. Also, an abundance of data-driven and nonlinear control approaches exist in the literature, including model-free adaptive control, extremum-seeking, feedback linearization, describing functions, sliding mode control, back-stepping, etc. However, Despite of vast progress that has been made in the development of nonlinear control systems, each of the aforementioned methods is just tailored to a specific class of problems. Moreover, these control approaches require huge mathematical and computational resources, or at least don't readily generalize to new applications. Currently there is no all-embracing framework for nonlinear control which is able to be generically applied to a wide class of potentially high-dimensional systems, as exists for linear systems. The French-born American mathematician Bernard Koopman(1900-1981) showed that it is possible to represent a nonlinear state-space equation in terms of an infinite-dimensional linear operator acting on a Hilbert space of measurement functions of the state of the system. That is, the finite dimensional nonlinear system can be described via linear equations with the cost of augmenting the number of the states possibly to the infinity. This poses new challenges. Obtaining finite-dimensional, matrix approximations of the Koopman operator is the focus of intense research efforts. Once the nonlinear system is expressed in a linear framework, it can enjoy the abundance of optimal estimation and control techniques that are available for linear systems and the ability to analytically predict the their future states. Therefore, one should search for a finite-dimensional approximation of the Koopman operator. This requires identifying a subspace spanned by a subset of eigenfunctions of the Koopman operator.

This report is organized as follows: section 1 introduces the Koopman operator. in section 2 I define the control objective which is to minimize a quadratic cost function. Section 3 deals with an example. The Koopman Quadratic Regulator is applied on a duffing oscillator. A Model-free design is also obtained and its result is compared to that of analytically obtained controller.

2. Problem formulation

Consider a continuous-time dynamical system, given by:

$$\frac{d}{dt}\mathbf{x}(t) = \mathbf{f}(\mathbf{x}), \quad \mathbf{x}(0) = \mathbf{x}_0 \quad (1)$$

The Koopman operator advances measurement functions of the state with the flow of the dynamics. We consider real-valued measurement functions $g : \mathbf{M} \rightarrow \mathbb{R}$, which are elements of an infinite-dimensional Hilbert space. The functions g are also commonly known as observables. They include every possible function of the states. For instance, $x_1 - x_3^2, x_1^2 \sin(x_1 x_3), x_1, x_2 + x_3 x_1$, etc are observables. The term 'observable' should not be confused with the unrelated observability from control theory. The Koopman operator $\mathcal{K}_t$ is an infinite-dimensional linear operator that acts on measurement functions g as:

$$\mathcal{K}_t g = g \circ \mathbf{F}_t \quad (2)$$

where $\circ$ is the composition operator. For a discrete-time system with timestep Δt , this becomes:

$$\mathcal{K}_{\Delta t} g(\mathbf{x}_k) = g(\mathbf{F}_{\Delta t}(\mathbf{x}_k)) = g(\mathbf{x}_{k+1}) \quad (3)$$

In other words, the Koopman operator defines an infinite-dimensional linear dynamical system that advances the observation of the state $g_k = g(\mathbf{x}_k)$ to the next time step.

The Koopman operator is linear, a property which is inherited from the linearity of the addition operation in function spaces

$$\begin{aligned} \mathcal{K}_t(\alpha_1 g_1(\mathbf{x}) + \alpha_2 g_2(\mathbf{x})) &= \alpha_1 g_1(\mathbf{F}_t(\mathbf{x})) + \alpha_2 g_2(\mathbf{F}_t(\mathbf{x})) \\ &= \alpha_1 \mathcal{K}_t g_1(\mathbf{x}) + \alpha_2 \mathcal{K}_t g_2(\mathbf{x}) \end{aligned} \quad (4)$$

The linear nature of Koopman operator allows us to define the eigenfunctions and eigenvalues of this operator as follows.

2.1. Koopman eigenfunctions

In order to address the infinite-dimension of the new states, instead of capturing the evolution of all of the observables in a Hilbert space, the applied Koopman analysis approximates the evolution on a subspace spanned by a finite set of measurement functions. It is possible to obtain a finite dimensional matrix representation of the Koopman operator by restricting it to an invariant subspace. This is conceptually similar to eigenvectors in the vector-space, where the eigenvector directions remained unchanged when passed to a matrix. Similarly, The eigenfunction of Koopman operator is a function $\phi_\lambda \in \mathcal{O}$ that satisfies:

$$\frac{d}{dt}\phi(\mathbf{x}) = \mathcal{K}\phi(\mathbf{x}) = \lambda\phi(\mathbf{x}) \quad (5)$$

Applying the chain rule to the time derivative of the Koopman eigenfunction $\phi(\mathbf{x})$ yields

$$\frac{d}{dt}\phi(\mathbf{x}) = \nabla\phi(\mathbf{x}) \cdot \dot{\mathbf{x}} = \nabla\phi(\mathbf{x}) \cdot \mathbf{f}(\mathbf{x}) \quad (6)$$

Combined with (5), this results in a partial differential equation (PDE) for the eigenfunction $\phi(\mathbf{x})$:

$$\nabla\phi(\mathbf{x}) \cdot \mathbf{f}(\mathbf{x}) = \lambda\phi(\mathbf{x}) \quad (7)$$

Depending on the system, it is possible to find the closed finite set of the eigenfunctions by solving the above PDE. Otherwise, the system can not be described with a finite number of eigenfunctions. In this situation, one can consider some candidate functions and then try to adjust the coefficients by solving an optimization problem. This procedure is followed in Extended Dynamic Mode Decomposition(EDMD) and Sparse Identification of Nonlinear Dynamics(SINDy) algorithms.

Let

$$\Phi(x) = [\phi_1(x), \dots, \phi_N(x)]^\top \quad (8)$$

comprise the first N dominant eigenfunctions with ϕ_j associated with eigenvalue λ_j . Consider now the transformation as $\hat{\Phi} : \mathbb{R}^n \rightarrow \mathbb{R}^N$ as:

$$z = \Phi(x) \quad (9)$$

Subsequently, the system equation takes the following form in this new coordinates:

$$\dot{z} = \Lambda z + \frac{\partial \hat{\Phi}}{\partial x} G(x) u \quad (10)$$

where Λ is a block diagonal matrix containing the eigenvalues of the Koopman matrix.

2.2. Control Objective

The control objective is to minimize the quadratic cost function:

$$J(\mathbf{x}, \mathbf{u}) = \int_0^\infty x^\top(t) \bar{Q} x(t) + R u^2(t) dt \quad (11)$$

Analogously, we can define the equivalent cost function in the space of Koopman eigenfunctions. The control objective then becomes:

$$J(\Phi, \mathbf{u}) = \frac{1}{2} \int_0^\infty \Phi^T(\mathbf{x}(t)) \mathbf{Q}_\Phi \Phi(\mathbf{x}(t)) + \mathbf{u}^T(t) \mathbf{R} \mathbf{u}(t) dt \quad (12)$$

For this cost function to be equivalent to the cost in the original state $\mathbf{x}$, a modified weight matrix may be considered such that $\Phi^T \mathbf{Q}_\Phi \Phi \approx \mathbf{x}^T \mathbf{Q} \mathbf{x}$. Both of the weight matrices are symmetric and positive definite. The LQR controller can be readily obtained via MATLAB 'lqr' command. As this is a familiar control approach we do not go through its details and directly head towards its application on the model.

3. Duffing Oscillator

3.1. System Definition

Certain damped or forced oscillators are described by the Duffing equation:

$$\dot{\mathbf{x}} = \begin{bmatrix} x_2 \\ x_1 - x_1^3 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad (13)$$

The system is steered towards the energy level $\mathcal{E} = 0$, which corresponds to the separatrix cycle yielding a periodic solution. The origin is a saddle point leading to a homoclinic orbit defined by $x_2^* = \pm x_1^* \sqrt{1 - (x_1^*)^2/2}$ for $x_1^* \leq \pm\sqrt{2}$. The state-trajectories of the system are delineated in the Fig. 2. It is possible to stabilize the center fixed points by commanding a lower reference energy. In fact, instead of commanding the system to settle on a reference point, the command is now to settle on a reference energy level.

Hamiltonians represent the energy orbits. Conserved quantities, such as the Hamiltonian $\mathcal{H}$, are Koopman eigenfunctions associated with the eigenvalue $\Lambda = 0$. Hamiltonian systems represent a special class of systems for which we can easily discover a Koopman eigenfunction directly from data. In other words, the hamiltonian belongs to the nullspace of the Koopman operator. Therefore, it can be obtained via the following PDE:

$$\frac{d}{dt} \mathbf{q} = \frac{\partial \mathcal{H}}{\partial \mathbf{p}}, \quad \frac{d}{dt} \mathbf{p} = -\frac{\partial \mathcal{H}}{\partial \mathbf{q}} \quad (14)$$

Thus, stabilization and tracking in state-space can be translated to stabilization of energy level. For the Duffing Oscillator, we consider the problem of reaching to the zero-energy orbit. Once the trajectory reaches this orbit, it remains there forever. For the Duffing oscillator, the Hamiltonian is:

$$\mathcal{H} = \frac{1}{2} x_2^2 - \frac{1}{2} x_1^2 + \frac{1}{4} x_1^4 \quad (15)$$

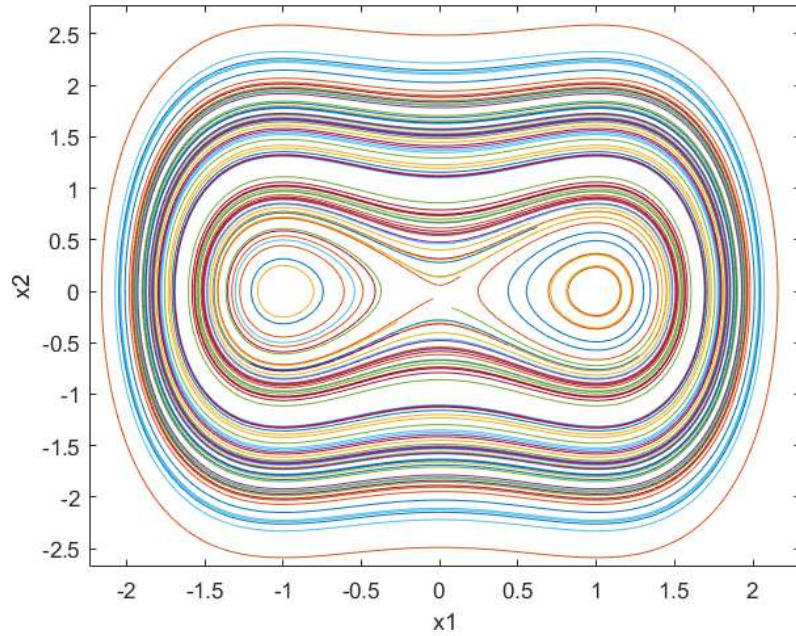


Fig. 1. Duffing Oscillator System

Considering this special eigenfunction $\Phi = \mathcal{H}$ whose eigenvalue equals zero, one can obtain the appropriate input matrix of the linear system:

$$\frac{d}{dt}\Phi = 0 \cdot \Phi + \nabla_{\mathbf{x}}\Phi \cdot \mathbf{B}\mathbf{u} := \mathbf{B}_{\Phi}\mathbf{u} \quad (16)$$

3.2. Data-driven identification of discrete-time eigenfunctions

The system is analyzed with computer, so it is discrete. In discrete time, an eigenfunction evaluated at a number of data points in $\mathbf{X}$ will satisfy:

$$[\lambda\Phi(\mathbf{x}_1) \dots \lambda\Phi(\mathbf{x}_m)]^T = [\Phi(\mathbf{x}_2) \dots \Phi(\mathbf{x}_{m+1})]^T \quad (17)$$

We may develop the control directly for the eigenfunction equation

$$(\lambda\Phi(\mathbf{X}) - \Phi(\mathbf{X}'))\xi = \mathbf{0} \quad (18)$$

where $\mathbf{X}' = [\mathbf{x}_2 \ \mathbf{x}_3 \ \dots \ \mathbf{x}_{m+1}]$ is a time-shifted matrix. This formalism directly identifies the functional representation of a Koopman eigenfunction with eigenvalue λ . The Koopman operator can be obtained by applying the least-squares to the above equation.

$$\mathbf{K} = \Phi^\dagger \Phi' \quad (19)$$

where $\Phi^\dagger$ is pseudo-inverse of Φ . Then to reach sparse approximation of Koopman operator, one can set the matrix elements lower than a small value identically zero.

The simulation on the system on MATLAB reveals that the data-driven Koopman operator, approximates the Hamiltonian as:

$$\mathcal{H} = \frac{2}{3}x_2^2 - \frac{2}{3}x_1^2 + \frac{1}{3}x_1^4 \quad (20)$$

Although the Koopman Operator in the analytical solution and the data-driven one are rather different, their corresponding closed-loop system are similar. Moreover, Fig. 3 shows that there is meagre difference between their control expenditure.

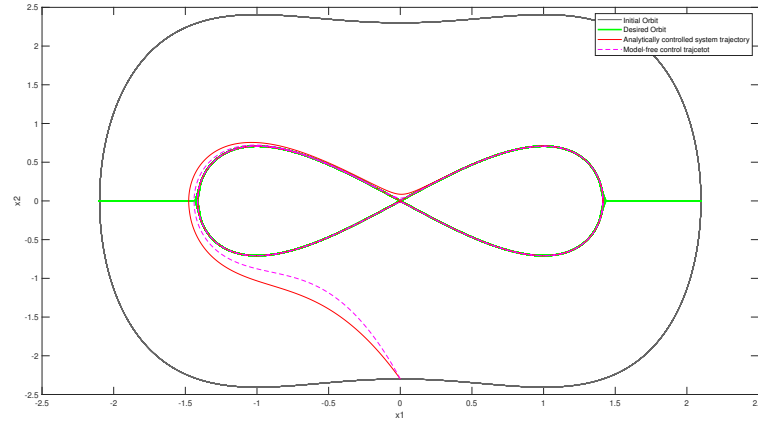


Fig. 2. Convergence of the trajectories

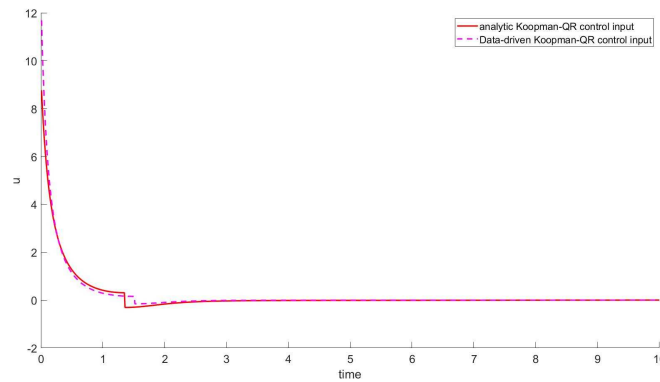


Fig. 3. control expenditure of Koopman Quadratic Regulated Duffing Oscillator for analytical and data-driven solution

3.3. Comparison to LQR

To evaluate the performance of the Koopman controller, I decided to compare it to a an LQR that is designed for the linearized approximation of the Duffing system. Fig. 4 and Fig. 5 delineate the results:

As is evident from the evolution of the states, it is palpably wrong to compare the cost function for these two systems. The reason is that the nonlinear one is designed to oscillate on the constant energy surface, while the linearly controlled one is just to asymptotically stabilize the system by converging to the origin.

The future work will amend this comparison to reach wise result.

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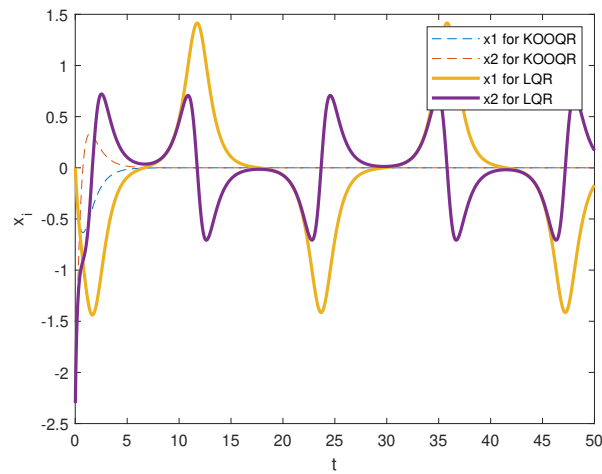


Fig. 4. Trajectories for the LQR vs KQR

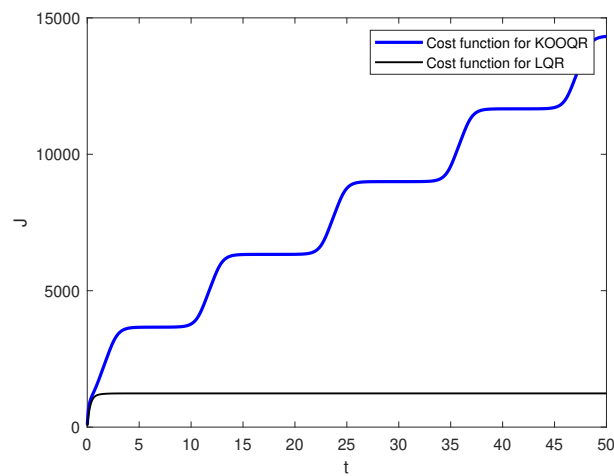


Fig. 5. Trajectories for the LQR vs KQR

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